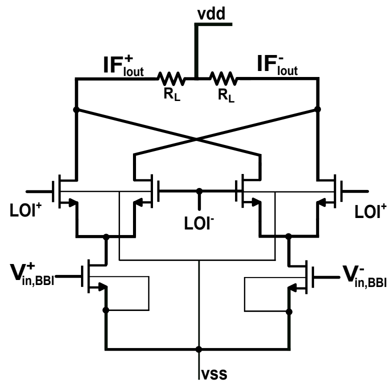
DOUBLE BALANCED ACTIVE MIXER ASSIGNMENT

Due Date: 05-04-2025



The goal of this project is to design a Double-Balanced Active Mixer (Gilbert Cell) in ADS using 40 nm CMOS.

System Specifications-

- Technology Node: 40 nm CMOS
- Nominal Supply Voltage: 1.1 V
- RF Input (Two-Tone): 2.40 GHz and 2.45 GHz
- LO Clock: 2.30 GHz (Sinusoidal)
- IF Targets: 100 MHz and 150 MHz
- Load Terminations: 50 Ω differential
- Output Capacitance: ≥ 20 fF

2. Minimum Viable Specifications

Metric	Target Value
Output Power	4 dBm
Suppression	≥ 35 dBc
IIP_3	≥ -10 dBm
Conversion Gain	≥ 1 dB

1. Determine the optimal device dimensions for the RF transconductance switching pair using the design methodology. Plot g_m/I_D versus V_{GS} and provide a justification for your selected region of operation.

$$V_{R,max} = V_{DD} - V_{X,min} = V_{DD} - \left[V_{GS1} - V_{TH1} + \left(1 + \frac{\sqrt{2}}{2}\right)\left(V_{GS2,3} - V_{TH}\right)\right]$$

then
Vth
then

2. Extract and plot the transconductance derivatives (gm_1, gm_2, gm_3) as a function DC bias voltage. Justify the theoretical optimal bias point required to maximize the third-order input intercept point (IIP3).
3. Plot the IIP3 versus the sweep of biasing of the gm-stage
4. What is the optimal DC bias condition and peak-to-peak voltage swing for the local oscillator (LO)? Provide a theoretical justification for these specific design choices.
5. Plot conversion gain as a function of the LO amplitude. Is there a transition boundary between soft switching and hard switching regions? Justify how your selected LO swing achieves an optimal balance
6. Plot the conversion gain versus the input power of the baseband while fixing the input of the LO drive at maximum linearity
7. Perform a two-tone Harmonic Balance. Plot the fundamental and IM3 IF output power versus RF input power and explicitly mark the point on the graph.
8. Plot the Input power of the baseband versus the output power and determine the 1 dB compression point of your design.

Bonus 1: Compare between resistive and inductive source degeneration to further enhance IIP3 of the gm-stage. Motivate your choice.

OR

BONUS 2: Simulate the noise figure of the Double Balanced Mixer and derive the theoretical noise figure. Do these values match?